



Sample Katz R01 Change in Research Direction Attachment

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NEW RESEARCH DIRECTIONS

My previous research examined auditory perception and cortical signal processing in humans, nonhuman primates (NHPs), and mice (non-disease models) primarily using auditory psychophysical tasks, acute single-unit recordings, and optogenetic manipulations. **My new research direction is to understand mechanisms linking sensory deprivation to cognitive impairment and hippocampal (HC) dysfunction, including Alzheimer's disease (AD) pathology.** My current proposal is to track the effects of hearing loss (HL) in a mouse model of AD, using behavioral tests of spatial memory, acute and chronic multisite recordings, and 3D histological imaging with iDISCO methodology. **Together, the questions raised in the current proposal and methods used to address them reflect a fundamental change in research direction.**

New conceptual directions. My prior research has aimed to understand how the central auditory pathway processes sounds and supports sound-guided behaviors, by addressing five general questions: (1) To what extent are sensory working memory processes shared between humans and NHPs? (2) How are sound features represented by single neurons and neuronal populations in NHP auditory cortex (ACtx)? (3) How do contextual variables such as motor activity, task engagement, and visual information modulate sound representations in NHP and mouse ACtx? (4) How do inhibitory interneuron subpopulations modulate sound representations in mouse ACtx? (5) How does partial sensory impairment (single-sided deafness) degrade cortical processing of the remaining sensory input? **Conceptually, these studies focused on sound processing and auditory cortical circuitry in healthy animals (not disease models).**

My new research direction is motivated by recent studies indicating that HL is a major modifiable risk factor for dementia including AD in humans and causes cognitive impairment (CI) reflecting HC dysfunction in rodents. These studies leave open two important questions I propose to address: first, does HL specifically accelerate AD pathology, and second, does HL disrupt neuronal activity patterns in HC? In Aim 1, I will determine whether HL accelerates AD pathology by surgically deafening mice carrying AD-related transgenes, and then examining hallmark AD symptoms at histological level (accumulations of amyloid β [A β] and phosphorylated tau [p-tau]) and physiological level (neuronal hyperactivity, diminished memory-related network oscillations (sharp-wave ripples [SWRs])). In Aim 2, I will examine a putative mechanism of HL-induced CI by longitudinally tracking neuronal activity in HC before and after HL, testing for changes including hyperactivity, SWR abundance, and functional interaction between HC and ACtx. **The current aims thus reflect a fundamentally new conceptual direction, as reflected by four general questions:** (1) Does HL accelerate accumulations of A β and p-tau in a mouse model of AD? (2) Does HL induce or exacerbate pathophysiological outcomes in HC including spontaneous hyperactivity and/or diminished SWRs? (3) Does HL induce changes in functional connectivity between HC and ACtx, and are these changes transient or long-lasting? (4) Are HL-induced changes in AD pathology, HC physiology, and/or CI dependent upon sex? **My new research direction thus focuses on neural pathology and activity in HC rather than sensory cortex, with the overall goal of understanding a disease process rather than normal function.**

New methodological directions. **My previous studies of auditory cognition and cortical processing relied on five main methods:** (1) auditory, visual, and/or tactile short-term memory tasks in humans and NHPs, (2) measuring responses to sound and other stimuli in NHP and mouse ACtx via acute extracellular recordings with single electrodes or arrays, (3) modeling sound encoding dynamics by single neurons and neuronal populations using spectrotemporal receptive field estimation techniques, (4) assessing the influence of acute manipulations of inhibitory interneuron subpopulations in mouse ACtx using optogenetic techniques, and (5) partial unilateral deafening through acute noise exposure. **My current proposal to examine the role of HL in HC dysfunction (including AD-specific pathology) will require six new methods:** (1) assessment of disease progression in a mouse model of AD, (2) histological assessment of AD pathology (A β and p-tau accumulations) using iDISCO methodology, which includes multiplexed immunolabeling, tissue clearing, light-sheet microscopy, and algorithmic whole-brain quantification, (3) induction of HL by cochlear ablation, (4) chronic recordings with high-density Neuropixels probes, (5) assessment of CI with a spatial learning and memory task, and (6) assessment of altered functional interaction between HC and ACtx using Granger causality analysis to quantify changes in the strength and directionality of influence between areas.

Comparison to prior work. The closest link to my prior work is that I have recent experience with a partial HL model (moderate single-sided HL through noise exposure). However, the proposed technique is somewhat different (complete HL via cochlear ablation vs. partial HL via noise exposure), and the motivating questions and goals of understanding disease progression are fundamentally different. Nevertheless, my familiarity with the HL literature first allowed me to identify major unresolved questions regarding the role of HL in AD and CI. My background knowledge, including dynamics of neural activity changes following HL, is thus relevant to the current proposal and directly influenced important aspects of the proposed experiments. Finally, my experience inducing and measuring HL mitigates one methodological risk associated with this new research direction.